EE4783: HW4

Prob. 1 (100 pts):

1) Translate the following C code into a Verilog code without pipelining.

```
x = 0;
for (i=0; i < 3; i++ ){
x = 2x + 1;
}
```

2) For your code in 1), find its throughput (bits/clock cycle), Latency (clock
cycles), and Timing (Critical path delay).

3) Now, pipeline your design in 1). Use 3 stages. List your circuit implementation.

4) For your code in 3), find its throughput (bits/clock cycle), Latency (clock cycles), and Timing (Critical path delay).